

Leukemia Cancer Detection in Microscopic Blood Samples Using Optimized Convolutional Neural Network

¹Gadi Himaja, Gadu Srinivasa Rao, Vinoothna More, Nitheesh Reddy Amedapu, Viyyapu Hamitha, Vishnuvardhan vinay vuppuluri, Dheeraj Chowdary Doppalapudi, Harshita Labhala, Karri Ajay
Department of Computer science and engineering
Gandhi Institute of Technology and Management,
Rushikonda, Visakhapatnam-530045, India
research23sn@gmail.com &himaja.himajagadi@gmail.com

Abstract— Leukaemia causes blood cell abnormalities quickly. Leukocytes injure bone marrow or blood and are followed by an increase in growing lymphocytes. Gelatinous tissue produces most of the body's blood. Thus, early and accurate cancer detection is essential for effective treatment that improves survival. Manually studying blood samples from microscopic pictures to identify this illness is slow, inaccurate, and time-consuming. This sickness is currently diagnosed manually by studying blood samples from microscopic images, which is arduous, slow, and imprecise. Furthermore, leukemic cells appear and act similarly to normal cells under a microscope, making identification more challenging. Though DL has developed advanced algorithms for detecting difficulties in recent decades, there is always room for improvement in performance, learning process, and efficacy. A hybrid approach using parallel swarm optimisation is more successful for identifying WBC leukaemia. To obtain high accuracy in picture classification, convolutional network hyperparameter tweaking is critical, but it involves significant computing effort. An automated leukaemia monitoring system for small blood samples was created utilising metaheuristics and parallel cat swarm optimisation (PCSO) to determine optimum convolutional neural network settings. Two public leukaemia blood sample databases, ALL IDB1 and ALL IDB2, are utilised. The method's resilience and effectiveness are shown by experimental findings and comparison with other cutting-edge algorithms under similar conditions.

Keywords— Convolutional Neural Networks, ALL IDB1 and ALL IDB2, Parallel Swarm Optimization

I. INTRODUCTION

U.S. cancer studies predict that 60,650 new cases of leukaemia will be detected in 2022, with 24,000 people losing their lives to the disease [1-3]. Bone marrow is the origin of leukaemia, which may then go to other organs via the circulation. In the very middle of each bone lies the marrow, the site of haemoglobin, platelets, and white blood cell (WBC) production. Lymphoid and myeloid cells are produced from stem cells, which are the original building blocks of blood cells [4]. Microscopic examination of blood smears on glass slides is the first step in identifying people with leukaemia. Medical diagnoses are made by specialists in their fields by examining microscopic slides. It takes a lot of time, effort, and expertise from the operator to complete this operation. Because manual diagnostic procedures are laborious and prone to error, precise imaging detection of blood malignancies is crucial for diagnosis, treatment

planning, and follow-up, particularly in clinical trials [4]. Consequently, a reliable automated method for the detection and diagnosis of blood malignancies is required. Medical imaging has recently detected blood cancer utilising image detection techniques. Additional detection approaches for leukaemia diagnosis include deep learning (DL) methodologies and machine learning (ML) algorithms [5]. Machine learning (ML) methods sometimes need preprocessing for object segmentation and augmentation before they can extract unique characteristics from the photos. In an effort to circumvent these limitations, recent studies have focused on diagnosing various ailments utilising DL neural networks [6]. One kind of artificial neural network, known as a Convolutional Neural Network (CNN), has shown to be very effective in computer vision tasks and has been able to reach expert levels in several domains, including computerised medical diagnosis. The enhanced PCSO is a more complex search strategy than the ones mentioned above; it uses data from previous rounds to choose future sets of hyperparameters [4].

CNN architecture and the proposed deep network's hyperparameters are chosen using the PCSO method. The suggested DL network's architecture and hyperparameters should be improved to increase detection performance and the capacity to discriminate between abnormal and healthy blood pictures. The following is a list of this study's main contributions:

- Creation of a new improved hybrid DL CNN for detecting ALL in photos of microscopic peripheral smears
- Creating a hybrid dataset by combining two publically open ALL datasets to provide more data to the input dataset.
- Determination of the learning parameters for the hybrid CNN using the PCSO method.
- Compare the optimised model to a an inefficient model in order to ascertain whether the suggested technique is helpful in boosting performance.
- Performance evaluation of the top network model and the current best practices.

II. RELATED WORKS

Several methods for automated leukaemia detection using microscopy images have been published in the scientific literature. These strategies include standard DL algorithms and ML classifiers. There are a number of systems that use DL methods in conjunction with ensemble machine learning to detect cancer and leukaemia. In order to speed up the finding of this deadly illness, automated leukaemia diagnosis has become crucial. Leukaemia detection in blood smear pictures has been the subject of much study using a number of methodologies. After extracting features from coloured microscopic photos using K-means, Thanh et.al. developed a method for classifying acute leukaemia blast cells based on their morphological properties. Emadi et al. developed a Hybrid Fuzzy C-Means algorithm for leukaemia diagnosis that combines Cluster Centre Estimation with a Support Vector Machine classifier. The collected characteristics were classified into normal and malignant cells in the cell picture using an SVM algorithm. When combined with other traditional ML classifiers, they could distinguish between normal lymphocytes and ALL cells in white blood cell pictures. The texture, colour, morphological, wavelet, and statistical properties were extracted. For choosing these features, they employed three different hybrid supervised feature selection techniques. A new hybrid optimized CNN with a PCSO algorithm for detecting Acute Lymphoblastic Leukemia detection is proposed. The suggested CNN's design is optimised using the optimization algorithm, which is also used to identify the ideal set of network hyperparameters. In comparison to more contemporary methods, the usefulness of employing PCSO for learning parameter tuning on the proposed network's performance is also examined.

III. LEUKEMIA CANCER DETECTION

A deep CNN based learning technique is presented for leukemia detection from microscopic blood smear cell samples. Each stage in the approach is thoroughly explained in the subsections listed below:

This inquiry uses coloured blood microscopic pictures of blood smear cells from the ALL-IDB databases. There are two parts to the ALL-IDB dataset: ALL-IDB1 and ALL-IDB2. In this study, we enhance the data by merging the two subgroups into a hybrid dataset of blood smear pictures. The total number of pictures in the hybrid dataset for blood smear cells is 368. Out of them, 189 represent healthy cells and 179 represent ALL cells. Using RGB photos from the ALL-IDB databases, blood cancer may be diagnosed. Forming, augmenting, scaling, and dividing hybrid sets are all possible thanks to the preprocessing of the picture data. It has already been mentioned that the ALL-IDB1 and ALL-IDB2 are joined to form a hybrid dataset. This hybrid dataset then augments the input dataset with additional data.

A) CNN Architecture

Convolutional neural networks (CNNs), powerful DL tools, have seen widespread use in picture identification. In terms of performance, CNNs are superior than traditional neural networks. Feature extraction is carried out using convolutional layers in order to extract unique properties from input pictures. In order to perform the classification operation, a fully connected layer receives feature maps that are generated by convolutional layers. An entirely novel convolutional neural network (CNN) architecture is detailed in this paper. Convolutional neural networks are powerful DL techniques that have been used to train the model's coefficients. Image classification often makes use of automatic parameter learning. CNNs outperform traditional neural networks. Model hyperparameters, on the other hand, are the characteristics that regulate the entire training procedure.

B) Parallel Cat Swarm Optimization (PCSO)

The PCSO process is described as follows: The procedure starts with the creation of N pixels, separated into G groups of pixels.

Step 1: Produce N pixels in the image, disperse them at random over the M-dimensional image pixel solution bounded within the initial value's constraint ranges, and then group them at random into G groups. Create the new velocities values for every image dimension in the interim.

Step 2: Initialize each cat's flag value in step two. Choose a random number of pixels and put them in searching mode while the others are set to tracing mode according to MR, where MR $[0,1]$.

Step 3: Incorporate the pixel coordinates into the fitness function to determine each pixel's fitness value. Keep track of the coordinate $x_{best,d}$ and fitness solution of the pixel with the best accurate fitness value after calculating.

Step 4: Move the pixels in accordance with their flags, if the pixels are in searching mode, apply them to that process; if not, apply them to the parallel pixel tracing process of the image.

Step 5: Re-select $[N [1-MR]]$ pixels to move in the seeking mode of the image and (N,MR) pixels to move in the parallel tracing mode of the image. This will reinitialize each pixel's flag and segregate them into locations that indicate the seeking or the tracing.

Step 6: Check whether the number of iterations in the optimization algorithm reaches a maximum iteration, if satisfied, apply the data interchanging process.

Step 7: Verify the termination condition. If satisfied, output the coordinate values that depicts the best solution and quit if you are satisfied; otherwise, move on to Step 3.

IV. EXPERIMENTAL RESULTS

The outcomes of the constructed model under various experimental settings are compared and explained in this section. The suggested model is built using Python and the Keras DL framework, and simulations on Google Colab with NVIDIA GPU. Two datasets are used in the experimental setting, and performance is validated both separately and jointly. The optimal set of parameter settings are used to report the findings after trial and error was used to define

each parameter. Fig.1. shows the enlarged Leukemia detection outcomes from the dataset. Fig.2. shows the Leukemia detection outcomes from the dataset. Fig.3. shows the Results of Augmented data on blood samples. The augmented data with the training samples are depicted. Fig.4. shows the Accuracy of proposed method with the compared Algorithms. It is observed that the accuracy is very high while comparing with the other approaches. The values marked with green color graph represents the accuracy of our approach. Fig.5. shows the Entropy Loss of proposed method with the compared Algorithms. It is observed that the entropy loss is very less while comparing with the other approaches. The values marked with green color graph represents the loss of our approach.

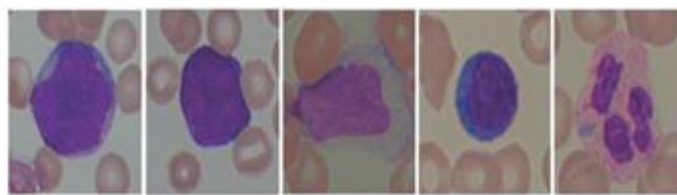


Fig. 1 Enlarged Leukemia detection outcomes from the dataset

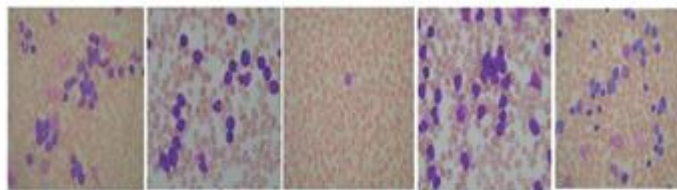


Fig. 2 Leukemia detection outcomes from the dataset

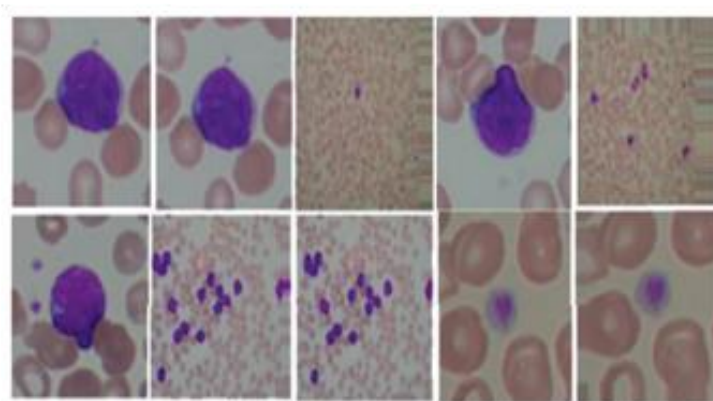


Fig. 3 Results of Augmented data on blood samples

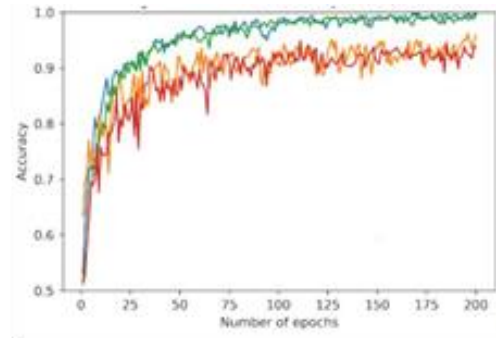


Fig. 4 Accuracy of the training samples with the compared Algorithms

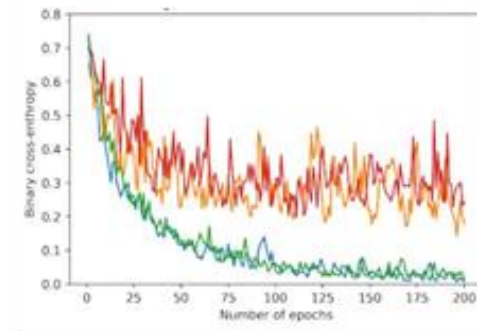


Fig. 5 Entropy Loss with the compared Algorithms

Blood cancer is detected using RGB images from the ALL-IDB datasets. The image data is preprocessed for hybrid set formation, augmentation, scaling, and splitting. As noted, the ALL-IDB1 and ALL-IDB2 are combined to create a hybrid dataset, which expands the input dataset by a data augmentation approach. DL networks generalization abilities and overfitting have been found to be reduced and improved by dataset augmentation, along with classification performance. The enhanced dataset consisted of 2208 photos in total, 1134 photographs of healthy people, and 1074 images of people with ALL diseases. In this paper, the suggested deep neural network's hyperparameters and optimal architecture are determined using the parallel swarm optimization approach. At this stage, the objective function is developed, the network hyperparameters that need to be improved are identified, and optimization is carried out. The best model is chosen at this stage, and the holdout test set is used to validate it. This method achieves cooperation in some way by requiring the subpopulations to exchange information. To regulate the switching of pixels between sub images, we establish the parameter CH. Every CH iteration, the pixel swapping is applied once. There are three steps in the pixel exchange process. Initially progressively select a group of the sub-images and sort the individuals in this group according to their fitness values in order to replace the individuals whose fitness value is the lowest in the group. Next, choose a local best solution at random from an unpredictable group. Thirdly, the local best solution is

chosen to replace the individual with the lowest fitness value in the group.

V. CONCLUSION

In this approach, the diagnosis of blood cancer in microscopic blood images is reviewed, and a novel hybrid CNN is proposed. The PCSO method is used to optimise the optimum CNN design and its learning parameters. The PCSO method is used to select the best network parameters and minimise an objective function. The ideal hyperparameters for the PCSO method are the convolution block depth, momentum, and initial learning rate. The newly described CNN was trained and validated using a mixed dataset. The hybrid picture collection has been enhanced via data augmentation through image changes in order to boost detection efficiency. The PCSO optimization approach's best CNN model's precision, specificity, and sensitivity for ALL detection. The PCSO optimization approach's best CNN model's accuracy, specificity, and sensitivity for ALL detection were noted. After being tested on publicly available datasets, the recommended approach shows promising results when compared to the traditional DL model. The suggested approach was also examined on other subtypes of acute lymphocytic leukaemia.

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